

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 70%. We keep your highest score.

1. Suppose you roll a fair six-sided die until you get a 6. What is the expected number of rolls until the first 6?

1 / 1 point

- ☐ 4
- ☒ 6
- ☐ 3
- ☐ 12

Nice work

The expected number of rolls until the first success (probability $\frac{1}{6}$) is $\frac{1}{\frac{1}{6}} = \frac{1}{1/6} = 6$.

2. Consider tossing a biased coin with $P(H \text{ ends}) = 0.4$, what is the probability that the first Heads occurs on or before the 3rd toss?

1 / 1 point

- ☒ 0.784
- ☐ 0.568
- ☐ 0.682
- ☐ 0.828

Nice work

assume X to be the random variable for the first index of seeing Heads.

$$P(X \leq 3) = P(X = 1) + P(X = 2) + P(X = 3) = 0.4 + 0.6 \cdot 0.4 + 0.6^2 \cdot 0.4 = 0.4 + 0.24 + 0.144 = 0.784 \%$$

3. Assume that the random variables X and Y are independent. Which of the following CAN NOT be concluded?

1 / 1 point

- ☐ $E[X + Y] = E[X] + E[Y]$
- ☒ $\text{Var}[XY] = \text{Var}[X] \cdot \text{Var}[Y]$
- ☐ $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
- ☐ $E[XY] = E[X] \cdot E[Y]$

Nice work

This is not true. $\text{Var}[XY] = \text{Var}(X)\text{Var}(Y) + \text{Var}(X)(E(Y))^2 + \text{Var}(Y)(E(X))^2$.

4. Suppose you flip a biased coin until you get heads. Assume the probability of heads is p . What is the probability that the first heads appear on the 4th flip?

1 / 1 point

- ☐ $(1 - p)^3 p^2$
- ☐ $(1 - p) p^3$
- ☐ p^4
- ☒ $(1 - p)^3 p$

Nice work

The first three flips must be tail which has the probability of $(1 - p)^3$ and the last one has to be heads. So the total probability is $(1 - p)^3 p$

5. If you flip a fair coin 10 times, what is the expected number of heads?

1 / 1 point

- ☐ 2
- ☒ 5
- ☐ 10
- ☐ 7

Nice work

The expected number of heads in 10 flips with $p = 0.5$ is $n \cdot p = 10 \cdot 0.5 = 5$.

6. If you flip a biased coin 5 times with $P(\text{heads}) = p$, what is the probability of getting exactly 3 heads?

1 / 1 point

- ☐ $\binom{5}{3} p^3$
- ☒ $\binom{5}{2} p^3 (1 - p)^2$
- ☐ $\binom{5}{3} p^3 (1 - p)^3$
- ☐ $\binom{5}{3} (1 - p)^3$

Nice work

Note that $\binom{5}{3} = \binom{5}{2}$. The answer is $\binom{5}{3} p^3 (1 - p)^2 = \binom{5}{2} p^3 (1 - p)^2$

7. You test 10 light bulbs, each with a 10% chance of being defective. What is the variance of the number of defective bulbs?

1 / 1 point

- ☐ 0.1
- ☒ 0.9
- ☐ 9
- ☐ 1

Nice work

The variance is $n \cdot p \cdot (1 - p) = 10 \cdot 0.1 \cdot 0.9 = 0.9$